

PROCESS-OPTION-VALUE ($o, (O, E), (V_S, V_V, E_S, E_C)$)

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1  if  $\exists v \in V_V : \text{id}(v) = \text{hash}(o)$  then
2    | return  $(V_S, V_V, E_S, E_C)$ 
3  end
4
5   $v \leftarrow (\text{hash}(o), \text{val}(o))$ 
6   $V_{V'} \leftarrow V_V \cup \{v\}$ 
7  let  $o' \in O : (o', o) \in E$ 
8  let  $v' \in V_V : \text{id}(v') = \text{hash}(o')$ 
9   $s' \leftarrow \text{spec}(v')$ 
10 if  $\exists s \in V_S \mid (s', s) \in E_C \wedge \text{name}(s) = \text{name}(o)$  then
11   |  $E_{S'} \leftarrow E_S \cup \{(v, s)\}$ 
12   |  $V_{S'} \leftarrow V_S$ 
13   |  $E_{C'} \leftarrow E_C$ 
14 else
15   | if  $o = \text{root}(O, E)$  then
16     | if  $V_S = \emptyset$  then
17       |  $s \leftarrow (\text{name}(o), \text{type}(\text{val}(o)))$ 
18       |  $V_{S'} \leftarrow V_S \cup \{s\}$ 
19     | else
20       | let  $s \leftarrow \text{root}(V_S, E_C)$ 
21       |  $V_{S'} \leftarrow V_S$ 
22     | end
23     |  $E_{C'} \leftarrow E_C$ 
24   | else
25     |  $s \leftarrow (\text{name}(o), \text{type}(\text{val}(o)))$ 
26     |  $V_{S'} \leftarrow V_S \cup \{s\}$ 
27     |  $E_{C'} \leftarrow E_C \cup \{(s', s)\}$ 
28   | end
29   |  $E_{S'} \leftarrow E_S \cup \{(v, s)\}$ 
30 end
31 foreach  $o_c \in O \mid (o, o_c) \in E$  do
32   |  $(V_{S'}, V_{V'}, E_{S'}, E_{C'}) \leftarrow \text{PROCESS-OPTION-VALUE}(o_c, (O, E), (V_{S'}, V_{V'}, E_{S'}, E_{C'}))$ 
33   | let  $v_c \in V_{V'} \mid \text{id}(v_c) = \text{hash}(o_c)$ 
34   |  $E_{C'} \leftarrow E_C \cup \{(v, v_c)\}$ 
35 end
36 return  $(V_{S'}, V_{V'}, E_{S'}, E_{C'})$ 

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PROCESS-OPTION-VALUE ($o, (O, E), (V_S, V_V, E_S, E_C)$)

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1  if  $\exists v \in V_V : \text{id}(v) = \text{hash}(o)$  then
2    | return  $(V_S, V_V, E_S, E_C)$ 
3  end
4
5   $v \leftarrow (\text{hash}(o), \text{val}(o))$ 
6   $V_{V'} \leftarrow V_V \cup \{v\}$ 
7  if  $V_S \neq \emptyset \wedge o = \text{root}(O, E)$  then
8    | let  $s \leftarrow \text{root}(V_S, E_C)$ 
9    |  $E_{S'} \leftarrow E_S \setminus \{(v, s)\}$ 
10 else
11   | let  $o' \in O : (o', o) \in E$ 
12   | let  $v' \in V_V : \text{id}(v') = \text{hash}(o')$ 
13   |  $s' \leftarrow \text{spec}(v')$ 
14   | if  $\exists s \in V_S \mid (s', s) \in E_C \wedge \text{name}(s) = \text{name}(o)$  then
15     |  $E_{S'} \leftarrow E_S \cup \{(v, s)\}$ 
16     |  $V_{S'} \leftarrow V_S$ 
17     |  $E_{C'} \leftarrow E_C$ 
18   | else
19     |  $s \leftarrow (\text{name}(o), \text{type}(\text{val}(o)))$ 
20     |  $E_{S'} \leftarrow E_S \setminus \{(v, s)\}$ 
21     |  $V_{S'} \leftarrow V_S \cup \{s\}$ 
22     | if  $o \neq \text{root}(O, E)$  then
23       |  $E_{C'} \leftarrow E_C \cup \{(s', s)\}$ 
24     | else
25       |  $E_{C'} \leftarrow E_C$ 
26     | end
27   | end
28 end
29 foreach  $o_c \in O \mid (o, o_c) \in E$  do
30   |  $(V_{S'}, V_{V'}, E_{S'}, E_{C'}) \leftarrow \text{PROCESS-OPTION-VALUE}(o_c, (O, E), (V_{S'}, V_{V'}, E_{S'}, E_{C'}))$ 
31   | let  $v_c \in V_{V'} \mid \text{id}(v_c) = \text{hash}(o_c)$ 
32   |  $E_{C'} \leftarrow E_{C'} \cup \{(v, v_c)\}$ 
33 end
34 return  $(V_{S'}, V_{V'}, E_{S'}, E_{C'})$ 

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